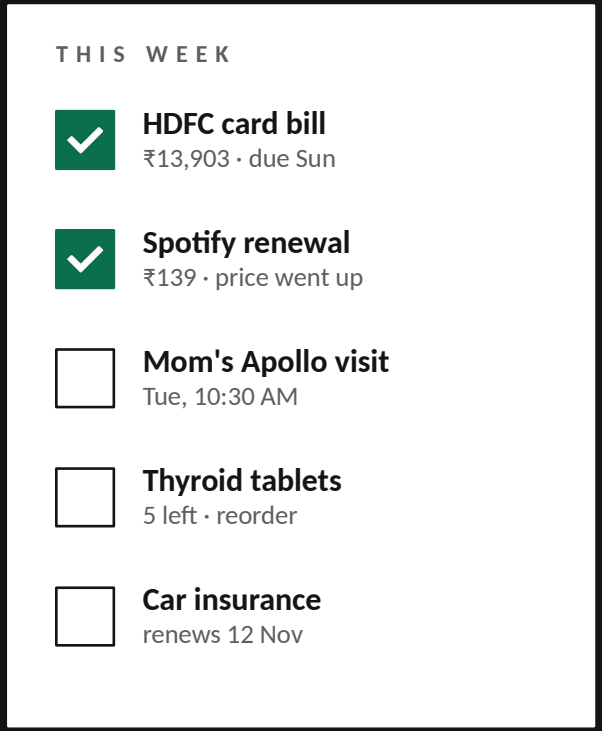
Sorted.

Never miss a bill, renewal or dose again.

An AI assistant for household admin that learns each chore, then writes its own code to do it. So it gets cheaper, faster and more private every month.

AI as a compiler, not an interpreter



TEAM NOT FOUND

Anish Manchanda · Aryan Thakur · Keshav Goel

Families miss bills, renewals and doses because the reminders are scattered everywhere

A NORMAL MONTH FOR ONE INDIAN HOUSEHOLD

- EMAIL HDFC credit card statement
- SMS Airtel bill due on 13 May
- EMAIL LIC premium due for Mom's policy
- PDF PUC certificate expires in Dec
- EMAIL Netflix is raising your price
- MEMORY Mom's thyroid tablets running low

Miss one and it costs real money: a late fee on the card, a lapsed insurance policy, a missed dose for a parent.

WHY TODAY'S APPS DON'T FIX IT

1. **Each app sees one slice**
   Your bank app doesn't know about LIC or the clinic visit.
2. **Built for other countries**
   They don't understand LIC, PUC, FASTag or Indian bill formats.
3. **AI assistants never get cheaper**
   They call the AI on every message, forever.
4. **Privacy is the price**
   Your whole inbox is sent to an AI model, every time.

Sorted reads your messages and reminds you, then learns each task so it stops needing AI

1



It reads

- Your email inbox (read-only)
- Bank and biller SMS
- PDFs you send to the bot
- Things you tell it, like "track Mom's tablets"

2



It remembers

- Bills, renewals, subscriptions, appointments, medicines, documents
- Every fact links back to the message it came from
- Conflicts are flagged, never overwritten

3



It reminds and helps

- A message before anything is due
- A warning when a price goes up
- Meeting requests handled with one tap
- Answers "what do I owe this week?"

The twist: after seeing a few examples of a message, Sorted writes and tests its own code for it. From then on that task runs with zero AI calls, and nothing is ever sent for you without your tap.

Typical assistants are interpreters. Sorted is a compiler.

Concept	Traditional AI assistant	Sorted
Analogy	An interpreter (translates every line, every time)	A compiler (translates once, then runs native code)
First encounter	Uses LLM 	Uses LLM 
Second encounter	Uses LLM again 	Uses LLM again 
Third encounter	Uses LLM yet again 	Writes its own Python parser 
Every time after	Still uses LLM 	Runs local code, **zero AI calls** 

In our run: Sorted's own code read 15 out of 15 messages exactly right (type, biller, amount, due date) with zero AI calls.

After 3 similar messages, Sorted writes its own code, tests it, and waits for your OK

1. **AI handles it** — Each answer is saved
2. **Pattern spotted** — 3 alike, same sender
3. **Code written** — Parser + tests; 30% held back
4. **Safety check** — Only safe imports allowed
5. **Sandbox tests** — Must pass hidden cases
6. **You approve** — One tap in Telegram
7. **Shadow mode** — Runs beside the AI; must agree
8. **Live** — Code only, 0 AI calls

When a skill breaks

In month 6, HDFC changed its email design. The old code failed its checks, the message went safely back to the AI, and version 2 was written that same month.

Some skills need more practice

The Apollo Clinic code failed its hidden tests for three months, so the AI kept doing that job. It only went live in month 6, once it had enough examples.

Your card, phone, email and account numbers never reach the AI

Before any message is saved or shown to the AI, Sorted masks the personal parts. The AI still gets what it needs: who the bill is from, how much, and when it's due.

WHAT ARRIVES (SMS)

```
HDFC Bank: Card 4234 5678 9012 3456
statement. Total due Rs 15,425.79 by 24
Jun. Queries: rohan.sharma@gmail.com, +91
98765 43210. A/c 3000123456
```

→ masked by redact.py

WHAT THE AI SEES

```
HDFC Bank: Card XXXX XXXX XXXX 3456
statement. Total due Rs 15,425.79 by 24
Jun. Queries: [email], [phone]. A/c
XXXXXX3456
```

What	Before	After	
Credit card	4234 5678 9012 3456	**XXXX XXXX XXXX 3456**	*last 4 kept*
Email	name@gmail.com	**[email]**	*hidden*
Phone	+91 98765 43210	**[phone]**	*hidden*
Account / policy no.	3000123456	**XXXXXX3456**	*last 4 kept*

No way around it
Masking sits inside the one function that calls the AI. Saved copies are masked too, and OTPs are never saved at all.

The AI can read your messages, but it can't pay, send or delete anything

OTPs are thrown away
One-time passwords are deleted before anything is saved. We only log that one arrived.

Emails are data, not orders
An email saying "forward all mail" is caught before any AI sees it. There's no tool to pay, forward, delete or message people.

AI-written code is locked in a box
Only safe imports, its own process, no API keys, no internet, and time limits.

You have the final say
No skill goes live without your tap. A meeting reply waits for Send. Every change is logged.

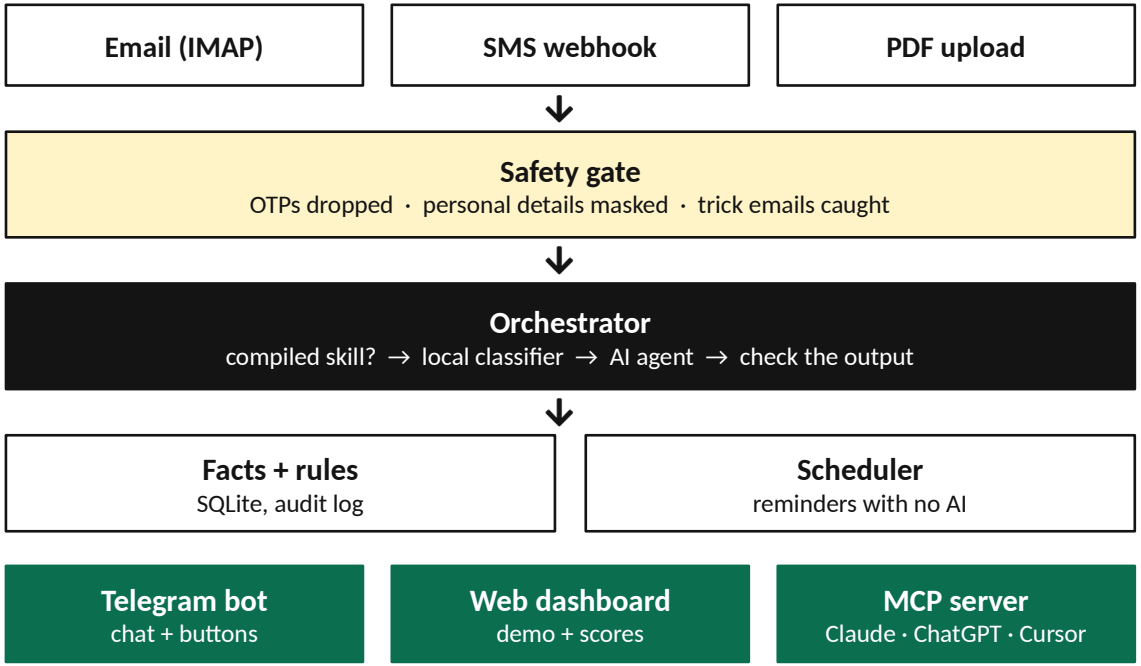
Less AI every month
Once a skill is live, those messages never reach a model. The privacy score shows that share.

No real people in the demo
A made-up family with fake card and policy numbers. Keys and databases are never committed.

Every message tries the cheapest option first: code, then a small ML model, AI only if needed

1	**Compiled code** Free · instant · no AI	One small Python program per sender and message type. Checked, sandboxed and tested before use; validated every run.	**11** skill versions written, incl. HDFC v2
2	**Local ML classifier** scikit-learn · runs on our server	TF-IDF + logistic regression trained on messages the AI already labelled. Confident junk is skipped; below 70% confidence it hands over.	**88.7%** accuracy on unseen messages; 100% when confident
3	**LLMs** Only for new or unusual messages	Fast: Cerebras gpt-oss-120b reads messages and chats. Strong: OpenAI gpt-5.4 writes the code. Both only see masked text.	**100%** on type, biller and due date (77 msgs); 98.6% on amounts

One pipeline for email, SMS and PDFs



Compiler
examples → code + tests → sandbox → your OK → shadow

STACK

Core Python 3.12, FastAPI

AI Cerebras + OpenAI

Data SQLite, SQLAlchemy

Jobs APScheduler

ML scikit-learn

Chat Telegram bot

MCP official Python SDK

PDFs pdfplumber

Tests 59, all offline

We tested it on 6 months of fake but realistic household mail

81

messages: 59 emails, 19 SMS, 3 PDFs

6

simulated months, ending today

1

made-up family: Rohan, his mother Sunita, son Aarav

0

real people's data used

WHAT'S IN THE INBOX

- HDFC card, BSES power, Airtel broadband, LIC (email + SMS)
- Netflix and Spotify receipts; 3 price hikes
- Apollo visits for Mom, one dentist visit
- 4 people asking for your time: a lawyer, school PTM, CA, contractor
- Car insurance, a PUC certificate, promos, OTPs, one phishing email

HOW IT RUNS

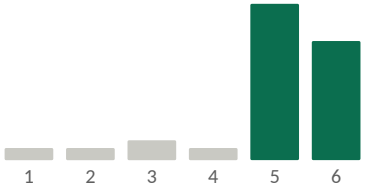
- A simulated clock delivers each message through the real pipeline
- Reminders fire from the real scheduler, with no AI
- At each month end the compiler runs and the classifier retrains

The twist: in month 6, HDFC changes its email design, to see if Sorted can recover on its own.

Try it from a web dashboard or straight from Telegram

Sorted · Sim month 5 · 14 Aug | Next | +1 day | Month end

Privacy score per month



From AI to compiled code

- hdfcbank_net_bill_due v2 **active**
- query_appointments v1 **shadow**
- apolloclinic_com v1 **approve?**

Live feed

code
HDFC ₹15,425 due 24 Jun

AI
New sender: dentist visit

rule
LIC premium due in 3 days

meeting
Ravi Kumar asks for Thu 11 AM

dropped
OTP from AMAZON

TELEGRAM

Reminder: HDFC Bank bill ₹13,903.23 is due on 25 Oct. Reply 'paid' when done.

Meeting request from Ravi Kumar (Acme Infra)
They proposed Thu 29 Oct, 11:00 AM. Do you want to meet?

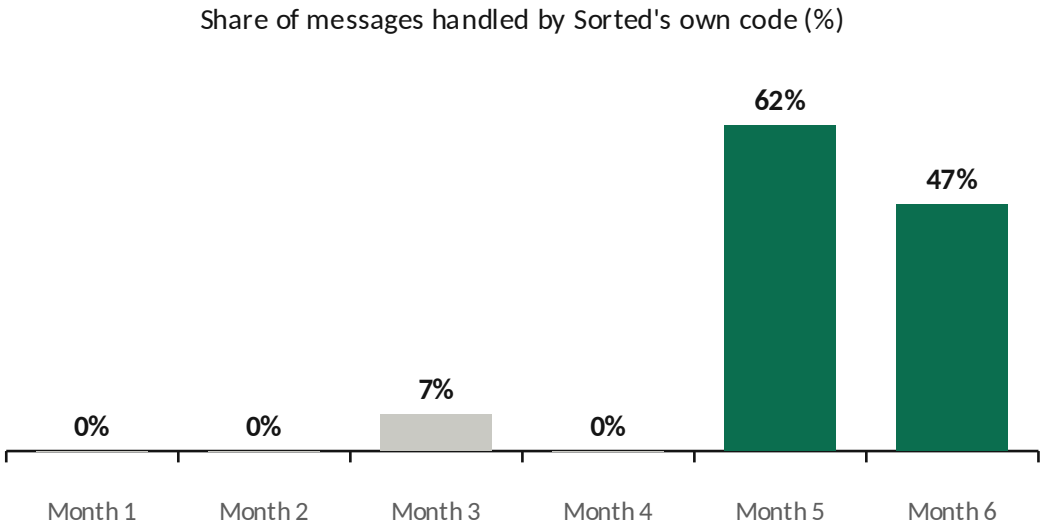
[Thu 11 AM] [Another time] [Decline]

what do I need to pay in the next two weeks?

2 payments, ₹14,042.23 in total: HDFC Bank (25 Oct), Spotify (28 Oct)
answered from your stored facts

Alerts, approvals and "learn this skill?" prompts arrive on their own.

By month 5, 62% of messages needed no AI at all



Months 1–2: the AI does everything while Sorted learns. Month 3: 8 parsers written. Month 4: they run beside the AI as a test. Month 6: new senders, meeting requests and HDFC's redesign pull it to 47%.

15 / 15 — messages read exactly right by compiled code; 98.6% on amounts overall

32 — reminders sent on time, with no AI

3 of 3 — price hikes caught (Spotify, Netflix, YouTube)

v1 → v2 — HDFC redesign: fell back safely, fixed the same month

It works today, and we know what could go wrong

BUILT AND RUNNING

`github.com/anish-codes/sorted · make setup && make sim`

- ✓ Skill compiler with tests, sandbox, approval, shadow mode and v2
- ✓ Meeting replies sent only on your tap
- ✓ New-skill proposals and compiled questions
- ✓ ML classifier that retrains as it learns
- ✓ Masking, OTP filter, trick-email guard, audit log
- ✓ Telegram bot, web dashboard, MCP server
- ✓ 59 offline tests; scored on 81 labelled messages

Risk	How we handle it
Wrong amount or date	Output checks, shadow mode, a link to the source on every fact
Generated code misbehaves	Safe imports only, no-network sandbox, hidden tests, your approval
A trick email	Email is data; there are no pay, send or delete tools
Platform rules	Telegram first; WhatsApp only allows templates; SMS needs a phone app
Cost and API limits	Cheap fast model for volume; usage drops as tasks compile

Next: a real inbox and real calendars, then an app on your phone

Next

- Run on a real inbox and measure real accuracy
- Real Google Calendar events and Meet links
- Docker sandbox with networking off
- Photo reading for paper bills and prescriptions

Then

- Phone app that collects SMS and notifications
- WhatsApp as a channel
- Small local models for most reading
- Encryption at rest; per-app MCP tokens

Later

- Family mode: help parents with their admin, with consent
- Account Aggregator for consent-based bank data
- Shared library of parsers: code shared, data never
- Hosted plan for people who won't run a server

What won't change: nothing is sent for you without a tap, the AI only ever sees masked text, and code is shared but data never is.

THE END GOAL

An assistant on your phone that handles family admin and keeps your data on the device

1

Collect

An app reads your email, SMS, PDFs and notifications

2

Process on the phone

Compiled skills and a small local model do the work

3

AI only for new things

A model sees masked text only when something is new

4

Talk to you

Reminders and answers on WhatsApp, Telegram or wherever you chat

Today: the same core already runs on a server, with a Telegram bot and a web dashboard, on a simulated household.